

POST-QUANTUM CRYPTOGRAPHY (PQC)

Executive Demo Guide, Client Value Proposition & Supported Technical Capabilities

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Good morning / afternoon. Today we are demonstrating the AI Cyber Audit Box Post-Quantum Cryptography (PQC) Engine -- an enterprise-grade automated solution built to discover, audit, and transition your organization's cryptographic infrastructure to NIST Quantum-Safe Standards.

Why Post-Quantum Cryptography Matters NOW ('Harvest Now, Decrypt Later'):
Adversaries are actively capturing encrypted network traffic, database backups, and sensitive communications TODAY. When Cryptographically Relevant Quantum Computers (CRQCs) emerge, algorithms like RSA-2048/4096, ECC P-256/384, and Diffie-Hellman will be broken instantly using Shor's Algorithm. Organizations must audit their Cryptographic Bill of Materials (CBOM) today to prevent retroactive data exposure.

NIST PQC Finalized Standards & Regulatory Compliance Frameworks:

- * FIPS 203 (ML-KEM / Module-Lattice Key Encapsulation): Primary quantum-safe key exchange standard.
- * FIPS 204 (ML-DSA / Module-Lattice Digital Signature): Primary quantum-safe digital signature standard.
- * FIPS 205 (SLH-DSA / Stateless Hash-Based Digital Signature): Backup signature standard for code/firmware signing.

2. 5-STEP LIVE DEMO FLOW (How to Perform Tomorrow's Walkthrough)

- Step 1: Uploading Evidence / Config Assets**

Upload client files (Nginx .conf, MySQL my.cnf, PostgreSQL sql.config, X.509 certificates, SSH configs, or binaries). Highlight multi-format support.
- Step 2: Instant Dual-Engine Scan**

Click 'RUN AUDIT SCAN'. Explain how the scanner parses both classical cipher suites and quantum readiness in sub-seconds with 100% determinism.
- Step 3: PQC Dashboard & Classification**

Show the live PQC Summary Panel. Explain the 3-tier classification: VULNERABLE (RSA/ECC), WEAK (SHA1/3DES), and SAFE (ML-KEM/PQC/AES-256).
- Step 4: Cryptographic Bill of Materials (CBOM)**

Navigate to the CBOM Register table. Show explicit key algorithms, key sizes, TLS protocol versions, and exposure tiers per asset.
- Step 5: 1-Click Executive PDF & Word Export**

Click 'Export PDF'. Present the audit report featuring executive summary, OEM readiness matrix, priority tiers, and actionable remediation steps.

3. CLIENT VALUE PROPOSITION & CORE SERVICES PROVIDED TO END USERS

1. Automated Cryptographic Bill of Materials (CBOM) Discovery

Provides complete visibility across enterprise infrastructure. Scans and inventories public/private key pairs, certificates, TLS/SSL protocol versions, SSH gateways, VPN tunnels, database encryption (TDE), KMS/HSM keys, and code signing certificates.

2. Quantum Vulnerability & Shor's Algorithm Risk Scoring

Evaluates every discovered algorithm against quantum attack vectors. Categorizes assets into VULNERABLE (quantum-breakable by Shor's algorithm: RSA, ECC, ECDSA, DH), WEAK (classically deprecated: MD5, SHA1, 3DES, RC4), and SAFE (quantum-resistant: AES-256, SHA-384, ML-KEM).

3. OEM & Vendor PQC Migration Readiness Matrix

Maps enterprise software vendors (MySQL, Nginx, Apache, OpenSSL, Windows Server, PostgreSQL, Java, Cloud Providers) against OEM PQC roadmap milestones to determine upgrade prerequisites.

4. Hybrid Cryptography & NIST Standard Migration Roadmap

Guides organizations in deploying hybrid key exchange mechanisms (e.g. X25519 + ML-KEM-768) and dual-signature schemes to ensure immediate post-quantum protection while maintaining backward compatibility with legacy clients.

5. Actionable Engineering Remediation & Governance Guidance

Delivers precise, copy-paste configuration snippets for system administrators and DevOps teams (e.g. ``ssl_ciphers``, ``tls_version``, OpenSSL 3.4+ PQC provider enablement) to eliminate vulnerabilities without downtime.

Quantum Status Classification Framework:

VULNERABLE

Quantum Breakable

RSA (any size), ECC (P-256/384), ECDSA, Diffie-Hellman

WEAK

Classically Insecure

3DES, DES, RC4, MD5, SHA-1, Single DES

SAFE

Quantum Resistant

AES-256, SHA-384, SHA-512, ML-KEM, ML-DSA, SLH-DSA

4. COMPREHENSIVE SUPPORTED FILE FORMATS & ASSET CAPABILITIES

The AI Cyber Audit Box PQC Engine includes native parsers capable of extracting cryptographic primitives, certificates, key specifications, and cipher suites across all enterprise file formats without requiring agent installation or cloud connection.

Asset Category	Supported File Extensions / Formats	Scanned Primitives & Extracted Metadata
Web & Proxy Servers	Nginx (.conf, .txt), Apache (httpd.conf, ssl.conf), HAProxy (haproxy.cfg), IIS (web.config)	ssl_ciphers, ssl_protocols, ssl_ecdh_curve, TLSv1.2, TLSv1.3, RSA, ECDSA, Curve25519, X25519
Database Encryption	MySQL / MariaDB (my.cnf, sql.config), PostgreSQL (postgresql.conf), Oracle DB (sqlnet.ora), MS SQL	require_secure_transport, tls_version, ssl-cert, ssl-key, ssl_ciphers, TDE master key algorithms
Certificates & Keys	PEM (.pem, .crt, .cer), DER (.der), PKCS#12 (.p12, .pfx), PKCS#8 (.key, .pkcs8), JKS (.jks)	X.509 OIDs, Signature Algorithm (sha256WithRSAEncryption), Public Key Type (RSA/ECDSA), Key Size (2048/4096)
SSH & Network VPN	OpenSSH (sshd_config, id_rsa, id_ed25519), OpenVPN (.ovpn), IPSec (ipsec.conf), WireGuard (wg0.conf)	KexAlgorithms, Ciphers, HostKeyAlgorithms, MACs, IKEv2 proposals, Ed25519, RSA-4096, AES-256-GCM
Code & Dependencies	Java (pom.xml, build.gradle), Python (requirements.txt, pyproject.toml, Node.js (package.json))	OpenSSL bindings, BouncyCastle PQC, PyCryptodome, cryptography lib, JS crypto imports
Binary & Containers	ELF Linux Binaries (/usr/bin), Windows Executables (.exe, .dll), JAR/WAR archives, Dockerfiles	Embedded OpenSSL string tables, libcrypto symbol exports, certificate bundles, hardcoded RSA keys

Dual-Engine Fast & Deep Parser Architecture:

- * Fast Rule Engine: Sub-millisecond deterministic regex and AST parsing for configuration files and certificates. Instantly flags quantum-vulnerable primitives with 100% precision and zero false positives.
- * Deep RAG Vector Engine: Analyzes unstructured audit evidence, architecture documents, policy PDFs, and unstandardized vendor documentation using ISO 27001 / NIST PQC RAG embeddings.

5. DEMO CHEAT SHEET & CLIENT Q&A (Common Questions & Answers)

Q1: Does the client need to provide private key files (.key) to run a scan?

ANSWER: NO. The PQC Engine parses public certificates (.crt/.pem), server configuration files (Nginx, Apache, MySQL), and cipher suite preferences. Private keys are NEVER required or exposed.

Q2: How fast is a typical scan across a bank or enterprise workspace?

ANSWER: Sub-second execution for configuration and certificate files. Multi-server batch audits complete in under 5 seconds.

Q3: Does the tool support NIST FIPS 203 (ML-KEM) and FIPS 204 (ML-DSA)?

ANSWER: YES. All discovered ciphers are directly benchmarked against NIST FIPS 203 (ML-KEM), FIPS 204 (ML-DSA), and FIPS 205 (SLH-DSA) standards.

Q4: Can this be deployed fully On-Premise in an Air-Gapped environment?

ANSWER: YES. The AI Cyber Audit Box is 100% self-contained with no external cloud API dependencies, ensuring complete data sovereignty for banking and defense clients.

6. KEY COMPETITIVE DIFFERENTIATORS (Why Choose Our PQC Engine)

- * 1. Instant Automated CBOM Generation -- Replaces weeks of manual spreadsheet audits with 1-click discovery.
- * 2. Dual PDF & Word Report Export -- One-click executive and engineering audit reports formatted for RFP compliance.
- * 3. Zero False Positives -- Dual deterministic rule engine + RAG verification ensures zero hallucinated gaps.
- * 4. OEM Migration Prerequisite Mapping -- Tells clients exactly which software versions (OpenSSL 3.4, Nginx 1.25) are required for PQC.

AI CYBER AUDIT BOX -- POST-QUANTUM CRYPTOGRAPHY ENGINE DEMO READY